# CLEANSCAN

User Guide

Software for restoring digitized film footage, designed for real-time preview, fine-tuning AviSynth+ processing, and high-quality encoding output.

**Trial version — encoding limited to 30 seconds per clip**

Publisher: **ScanFilm SNC**
Website: www.scanfilm.ch

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# 1. Overview

CleanScan is a restoration software for digitized analog film footage (8mm, Super 8, 9.5mm, 16mm).

It can reduce film grain, remove dust and scratches, correct brightness, contrast and colors, crop the image, and enhance sharpness.

The software generates an optimized AviSynth+ script and offers real-time preview through an integrated video player (libmpv). Final encoding is handled by FFmpeg.

# 2. Requirements and Installation

## System requirements

- Windows 10 or Windows 11 (64-bit)
- 4 GB RAM minimum (8 GB recommended)
- Sufficient disk space for source and output files

## Components used

- **AviSynth+ 64-bit** must be installed. If missing, CleanScan detects it at startup and offers a download link. A bundled version is used as fallback.
- **FFmpeg** is included with the application in the `Plugins/ffmpeg/` folder.
- **mpv** is included with the application in the `mpv/` folder.

## Installation

No additional installation required. Simply extract the archive and launch `CleanScan.exe`.

# 3. General Interface

The interface is organized into three main areas.

## Upper area

- **Menu bar:** Infos, Preset, Settings, Feedback, Language, About
- **Source bar:** source file selection, image mode, FPS
- **Clip tabs:** each loaded clip appears as a tab; the + button adds a clip
- **Transport bar:** playback controls, progress bar, and Record button

## Central area

Contains the mpv video player, displaying the AviSynth script preview in real time.

## Lower area

Contains the filter panel, split into two columns separated by a resizable splitter. The panel height is also adjustable.

# 4. Source Selection

## Video file

You can drag and drop a video file into the playback area, or click the Source field.
Supported formats: `.avi` `.mp4` `.mov` `.mkv` `.wmv` `.m4v` `.mpeg` `.mpg` `.webm`

## Image sequence

Enable image mode, then specify the folder path in the **Img** field. Supported formats: `.tif` `.tiff` `.jpg` `.jpeg` `.png` `.bmp`. Then specify the first and last image via the **Start** and **End** fields.

## Frame rate (FPS)

| Film format | Typical frame rates |
|---|---|
| 8mm | 16 fps or 18 fps |
| Super 8 | 18 fps or 24 fps |
| 16mm | 8 fps or 24 fps |

## Multi-clips

You can load multiple files simultaneously. Each clip has its own filter settings. Add them via multiple drag-and-drop or with the **+** button.

# 5. Video Player and Navigation

The built-in player displays the AviSynth script result in real time. Any parameter change automatically regenerates the script and updates the preview.

## Keyboard shortcuts

| Shortcut | Action |
|---|---|
| Space | Play / Pause |
| Left arrow | Previous frame |
| Right arrow | Next frame |
| Ctrl + Left | Go to beginning |
| Ctrl + Right | Go to end |

## Transport bar

| Button | Function |
|---|---|
| \|< | Beginning of clip |
| << | Previous frame |
| > | Play / Pause |
| >> | Next frame |
| >\| | End of clip |
| **1x** | Speed (0.25x / 0.5x / 1x) |
| **1/2** | Half resolution (faster) |

The progress bar displays the current position and total duration. Click or drag the slider to navigate through the clip.

The dropdown menu to the right of the transport bar lets you apply a filter preset to the active clip only.

# 6. Filter Panel

The lower panel is split into two columns: left column for activation buttons, right column for detailed parameters. Each filter has an ON/OFF button (green when active). An expansion button (>) displays detailed parameters. Hovering over parameters shows tooltips with recommended ranges.

## General options

| Parameter | Description |
|---|---|
| `preview` | AviSynth preview |
| `enable_flip_h` | Horizontal flip |
| `enable_flip_v` | Vertical flip |

## Crop

Removes pixels from the image edges.

| Parameter | Range | Default |
|---|---|---|
| `Crop_L` | 0 to 500 px | 0 |
| `Crop_T` | 0 to 500 px | 0 |
| `Crop_R` | 0 to 500 px | 0 |
| `Crop_B` | 0 to 500 px | 0 |

## Degrain (grain reduction)

Temporal grain reduction via MVTools2. Available presets: *low, standard, medium, strong, very strong*.

| Parameter | Range / Values | Default |
|---|---|---|
| `degrain_mode` | mdegrain1 / mdegrain2 / mdegrain3 / temporal | mdegrain2 |
| `degrain_thSAD` | 0 to 1000 | 350 |
| `degrain_thSADC` | 0 to 1000 | 250 |

| Parameter | Range / Values | Default |
|---|---|---|
| `degrain_blksize` | 8 / 16 / 32 | 16 |
| `degrain_overlap` | auto | blksize/2 |
| `degrain_pel` | 1 or 2 | 2 |
| `degrain_search` | 0 / 2 / 3 | 2 |
| `degrain_prefilter` | remgrain / blur / none | remgrain |

> **Notes:** mdegrain2 = 2 temporal references (recommended). mdegrain3 = 3 references (best quality, slower). mdegrain1 = 1 reference (faster). temporal = uses TemporalSoften (without MVTools2).

## Denoise (dust removal)

Temporal detection and removal of dust and scratches. Available presets: *low, standard, medium, strong, very strong*.

| Parameter | Range / Values | Default |
|---|---|---|
| `denoise_mode` | removedirtmc / removedirt | removedirtmc |
| `denoise_strength` | 1 to 24 | 10 |
| `denoise_dist` | 1 to 20 | 3 |
| `denoise_grey` | true / false | false |

> **Recommendation:** removedirtmc (with motion compensation) is recommended.

## Luma Levels (brightness levels)

| Parameter | Range | Default |
|---|---|---|
| `Lum_Bright` | -255 to 255 | 0 |
| `Lum_Contrast` | 0.1 to 3.0 | 1.05 |
| `Lum_Sat` | 0.1 to 3.0 | 1.10 |

| Parameter | Range | Default |
|---|---|---|
| `Lum_Hue` | -180 to 180 | 0 |
| `Lum_GammaY` | 0.1 to 3.0 | 1.30 |

> A GammaY value above 1 brightens the image; below 1 darkens it.

## GamMac (color correction)

| Parameter | Range | Default |
|---|---|---|
| `LockChan` | -3 to 2 | 1 (Green) |
| `LockVal` | 1 to 255 | 250 |
| `Scale` | 0 to 2 | 2 |
| `Th` | 0 to 1 | 0 |
| `HiTh` | 0 to 1 | 1 |
| `X, Y, W, H` | 0 to ... | 0 (full image) |
| `Omin` | 0 to 255 | 0 |
| `Omax` | 0 to 255 | 255 |
| `Show` | true / false | false |
| `Verbosity` | 0 to 6 | 0 |

> **LockChan values:** 0 = Red, 1 = Green, 2 = Blue, -1 = target value, -2 = average, -3 = median.

## Sharpen

Image sharpening. Available presets: *light, standard, medium, strong, very strong*.

| Parameter | Range / Values | Default |
|---|---|---|
| `Sharp_Mode` | simple / edge | simple |

| Parameter | Range / Values | Default |
|---|---|---|
| `Sharp_Strength` | 1 to 20 | 8 |
| `Sharp_Radius` | 0.5 to 5.0 | 1.5 |
| `Sharp_Threshold` | 0 to 100 | 0 |

> **simple** = global Sharpen() (effective, may amplify grain). **edge** = contours via Sobel mask + gamma (better preserves smooth areas).

# 7. Presets

The **Preset** menu lets you save and restore complete filter configurations.

- **Save:** saves current settings under a name
- **Load:** applies an existing preset
- **Delete:** removes a preset

Source, film, img, start, end, and crop fields are not included in presets.

Storage location: `%AppData%\CleanScan\presets.json`

# 8. Encoding and Rendering

Click the **REC** button in the transport bar to open the encoding panel. It has two columns: on the left, the list of clips to encode; on the right, encoding parameters.

## Main steps

1. **Clip selection:** check the clips to encode. Use "Select all" for batch processing.
2. **Output settings:** choose the destination folder. Available disk space is displayed.
3. **Encoder choice:** select the codec suited for broadcast, archiving, or post-production.
4. **Container choice:** MKV, MP4, AVI, or MOV for video codecs.
5. **Advanced parameter settings:** quality, chroma, and output resolution.

## Available encoders

| Encoder | Type | Use case |
|---|---|---|
| **x264** | Lossy | Broadcast (H.264) |
| **x265** | Lossy | Broadcast (HEVC) |
| **FFV1** | Lossless | Professional archiving |
| **UT Video** | Lossless | Fast editing |
| **ProRes** | HQ Compression | Post-production |
| **TIFF seq.** | Lossless | Individual images |
| **PNG seq.** | Lossless | Individual images |

## Advanced parameters

| Parameter | Description |
|---|---|
| **CRF** | Constant quality. Typical values: 18 to 23. |
| **Bitrate** | Fixed rate in kbps. |
| **Chroma subsampling** | 4:2:0, 4:2:2, or 4:4:4. |

| Parameter | Description |
|---|---|
| **Output resolution** | Original, 1080p, 720p, 576p, 480p. |

## During encoding

- Video playback is automatically stopped
- The interface is locked, except the Stop button
- A progress bar shows the progress
- Closing the application interrupts the current encoding

The **"Shutdown after encoding"** option automatically shuts down the computer after successful encoding.

> Encoding uses a dedicated render script ( `ScriptRender.avs` ) with preview mode disabled, to ensure maximum final quality.

# 9. Application Settings

| Setting | Description |
|---|---|
| **Threads** | Number of AviSynth threads. Increase to use more CPU cores. |
| **Source loader** | Video decoding plugin used for opening files. |
| **Reset settings** | Resets all settings to defaults. |

## Available loaders

| Loader | Description |
|---|---|
| **Auto** | Automatic detection (default) |
| **FFMS2** | FFmpegSource2 (MP4 / MKV) |
| **L-SMASH** | LSmashSource (MP4 / MOV) |
| **LWLibav** | LWLibavVideoSource |

## Languages

Available languages: Français, English, Deutsch, Español. The AviSynth script and interface are regenerated in the selected language.

# 10. Script Preview

The **Infos** > **Script preview** menu opens a window displaying the full contents of the `ScriptUser.avs` file generated by CleanScan.

The **Reload** button re-reads the file from disk and updates the display.

This feature lets you:

- Verify the generated script
- Copy it for external use
- Precisely check the applied processing

# 11. Feedback

The **"Feedback"** menu opens a built-in contact form to send the developer a bug report, feature request, or general feedback with star rating.

The form is sent via the Internet. Name and email address are optional but recommended if you want a response.

# 12. Recommended Workflow

1. Open your source (drag-and-drop or Source field).
2. Navigate to a representative frame, neither too dark nor too bright.
3. Enable filters progressively: Crop, Degrain, Denoise, Luma, GamMac, then Sharpen.
4. Use presets for quick setup, then fine-tune manually.
5. Check the result across several passages of the film.
6. Save your configuration as a preset.
7. Open the Record panel and start the final encoding.

# 13. Tips and Shortcuts

- The mouse wheel adjusts numeric fields incrementally.
- Half-resolution mode (**1/2**) significantly speeds up preview on less powerful machines.
- All parameters are automatically saved to `ScriptUser.avs` on each change.
- Window position, size, and panel states are remembered between sessions.
- The Record panel is restored on startup if it was visible when closed.
- Tooltips show value ranges and usage tips.
- Dragging and dropping multiple files automatically creates a separate clip for each one.

# 14. Files and Locations

| File / Folder | Location |
|---|---|
| `ScriptUser.avs` | Executable folder |
| `ScriptMaster.xx.avs` | Executable folder |
| `ScriptRender.avs` | Executable folder |
| `presets.json` | `%AppData%\CleanScan` |
| `encoding_presets.json` | `%AppData%\CleanScan` |
| `window_settings.json` | `%AppData%\CleanScan` |
| `session.json` | `%AppData%\CleanScan` |
| `Plugins/` | Executable folder |
| `mpv/` | Executable folder |

> **Support:** use the built-in Feedback form in the application.